YAMLess — pos-elera-sales-compliance AKS 2026-11-11

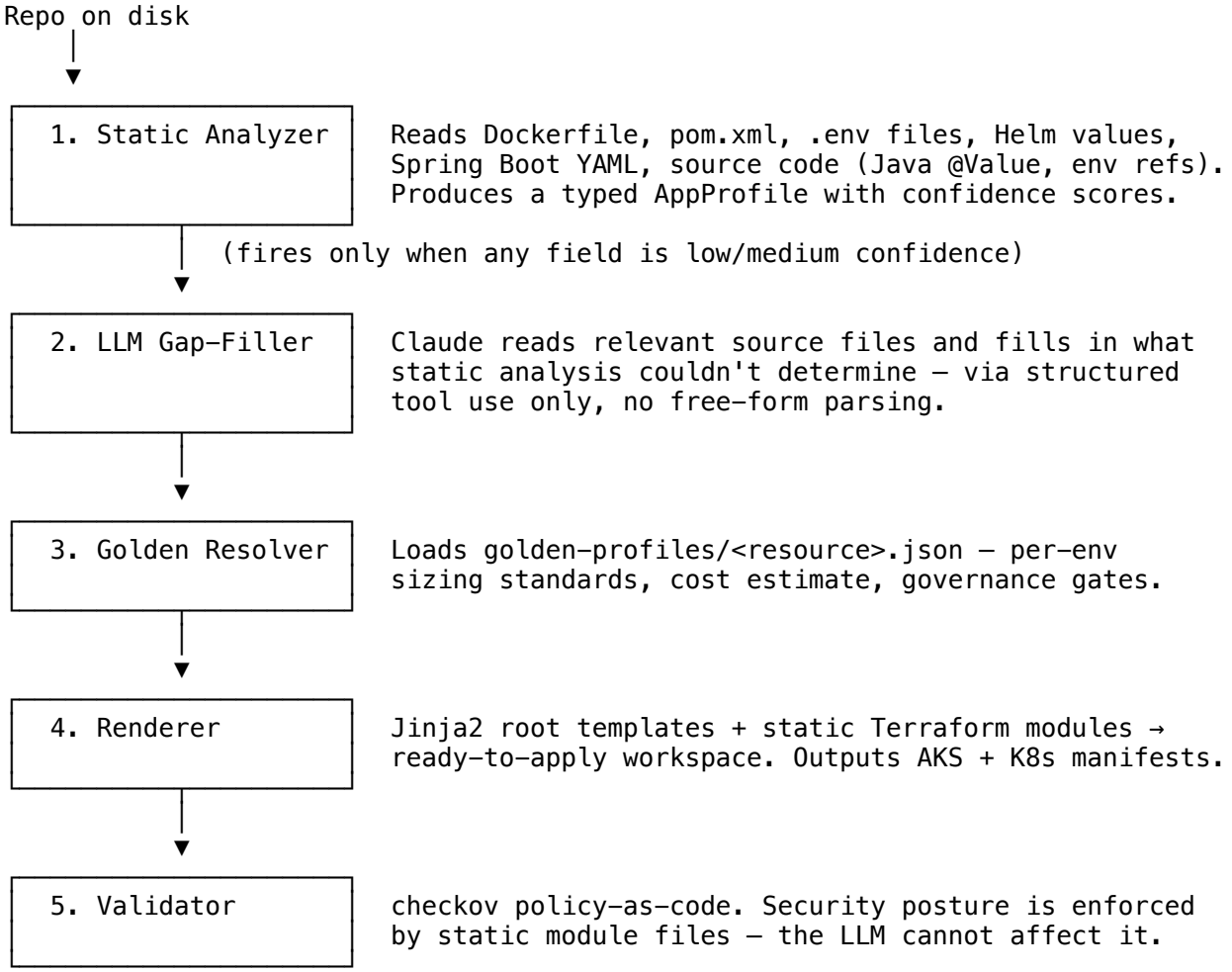
YAMLess: Generating Azure Infrastructure from Code

Case Study — krogertechnology/pos-elera-sales-compliance (all fixes through #36)

What is YAMLess?

YAMLess is an **Infrastructure from Code (IFC)** tool. Given a code repository, it produces a complete, policy-validated Terraform workspace — with no configuration files, no schema to fill in, and no IaC expertise required from the developer.

The pipeline has four stages:



What Changed Since v2

This run incorporates six fixes shipped across issues #28, #29, #31, #34, #35, and #36, all merged to main on 2026-11-11.

Issue	Fix	Effect on this repo
#28	Read `management.endpoints.web.base-path` from Spring Boot YAML; use `spring-boot-starter-actuator` as hint	`health_check_path: /health` → `/actuator/health`
#29	Extract `resources.limits/requests.memory` from Helm `values.yaml`; fallback to `-Xmx` inference	`mem_limit: null` → `1152Mi`; `mem_request: null` → `256Mi`
#31	Extract `replicaCount` / `autoscaling.minReplicas` from Helm `values.yaml`	No change — `replicaCount` absent in this repo's Helm
#34	Added `_SECRET_DENYLIST_RE` to filter `GF_`, `GRAFANA_`, `SWAGGER_`, `SPRINGDOC_`, `ACTUATOR_` prefixed vars	3 vars removed: `GF_SECURITY_ADMIN_PASSWORD`, `SWAGGER_URL`, `SWAGGER_CONFIG_URL`
#35	Added `azurerm_cosmosdb_account` module (MongoDB kind) for AKS and Container Apps renderers; `MONGODB_URI` / `SPRING_DATA_MONGODB_URI` marked `infra_managed`	Cosmos DB module now generated; 2 redundant KV secrets eliminated
#36	Emit warning when `len(secrets) > 30`; surface in CLI and profile	⚠️ "High secret count (39)" warning emitted

Infrastructure delta

Resource	v2	This run
AKS Cluster	✓	✓
Redis Cache (P1)	✓	✓
Service Bus	✓	✓
Key Vault	✓	✓
Cosmos DB for MongoDB	**✗ missing**	**✓ generated**

Applying YAMLess to pos-elera-sales-compliance

pos-elera-sales-compliance is a Kroger Technology Java Spring Boot microservice responsible for POS (point-of-sale) sales compliance. It integrates with the ELERA POS platform, Ping Identity for OAuth2/OIDC SSO, Kount for fraud detection, MongoDB for persistence, and RabbitMQ for messaging.

Command run

```
yamless generate https://github.com/krogertechnology/pos-elera-sales-compliance \
  --no-llm \
```

```
  --target aks \
  --out-dir ~/yamless/outputs/pos-elera-sales-compliance-aks-2026-11-11
```

Profile detected

Field	Detected value	Confidence	Δ vs v2
Runtime	`java`	high	unchanged
Runtime version	`25`	high	unchanged
Port	`8080`	—	unchanged
Shape	`http-service`	high	unchanged
Health check path	`/actuator/health`	—	**`/health`** → **`/actuator/health`** ← fix #28
Memory limit	`1152Mi`	—	**`null`** → **`1152Mi`** ← fix #29
Memory request	`256Mi`	—	**`null`** → **`256Mi`** ← fix #29
Database	`mongodb`	high	unchanged
Cache	`true`	high	unchanged
Queue	`true`	high	unchanged
Kafka	`false`	—	unchanged
Storage	`false`	—	unchanged
Secrets	39 detected	high	**42** → **39** ← fix #34
Min replicas	`1`	—	unchanged
Max replicas	`10`	—	unchanged
Ingress	`external: true`, port `8080`	high	unchanged
Scale trigger	`http`	high	unchanged

Output generated

```
pos-elera-sales-compliance-aks-2026-11-11/
├── profile.json
├── terraform/
│   ├── main.tf
│   ├── variables.tf
│   ├── outputs.tf
│   ├── providers.tf
│   ├── backend.tf
│   ├── governance.tf
│   └── modules/
│       ├── aks_cluster/
│       ├── aks_workload_identity/
│       ├── keyvault/
│       ├── cosmos/         ← NEW: Cosmos DB for MongoDB
│       ├── redis/
│       ├── servicebus/
│       ├── vnet/
│       └── (postgres, mysql, storage, event_hubs — generated but unused)
└── k8s/
    ├── namespace.yaml
    ├── service-account.yaml
    ├── secret-provider-class.yaml
    ├── deployment.yaml
    ├── service.yaml
```

```
    ├── ingress.yaml
    ├── hpa.yaml
    └── network-policy.yaml
```

38 .tf files + 8 K8s manifests written.

Cosmos DB module (new in this run)

The generated `module "cosmos"` block in `main.tf`, sized by `golden-profiles/azure-cosmos-db.json` for `production`:

```
module "cosmos" {
  source              = "./modules/cosmos"
  app_name            = local.app_name
  environment         = local.environment
  location            = local.location
  resource_group_name = azurerm_resource_group.main.name
  tags                = local.tags
  keyvault_id         = module.keyvault.id
  keyvault_rbac_ready = module.keyvault
  consistency_level   = "BoundedStaleness"
  throughput_mode     = "autoscale"
  max_autoscale_ru    = 10000
  server_version      = "4.2"
}
```

Golden standards enforced: `kind = "MongoDB"`, `EnableMongo` capability, public network access disabled, MongoDB API v4.2, writes `mongodb-url` to Key Vault — no connection strings in plaintext.

Deployment manifest (key fields fixed in this run)

```yaml
containers:
  - name: pos-elera-sales-compliance
    ports:
      - containerPort: 8080
    livenessProbe:
      httpGet:
        path: /actuator/health     # was /health in v2
        port: 8080
      initialDelaySeconds: 30
      periodSeconds: 10
      failureThreshold: 3
    readinessProbe:
      httpGet:
        path: /actuator/health     # was /health in v2
        port: 8080
      initialDelaySeconds: 10
      periodSeconds: 5
      failureThreshold: 3
    resources:
      requests:
        cpu: 250m
        memory: 256Mi              # was default in v2
      limits:
        cpu: 500m
        memory: 1152Mi             # was default 512Mi in v2
```

Cost estimate

Environment	Monthly	Annual	Δ vs v2
Dev	$170 – $245	$2,040 – $2,940	+$5–$30
Staging	$392 – $542	$4,704 – $6,504	+$50–$150
Production	$1,217 – $1,867	$14,604 – $22,404	+$200–$800

Production breakdown:

Resource	Cost/month	SKU	Note
AKS Cluster	$600	Standard	unchanged
Cosmos DB (MongoDB)	**$200 – $800**	**Autoscale 10,000 RU**	**new**
Redis Cache	$370	Premium P1	unchanged
Service Bus	$32 – $82	Standard	unchanged
Key Vault	$15	Premium	unchanged
Total	**$1,217 – $1,867**		+$200–$800 vs v2

The Cosmos DB range reflects autoscale RU consumption variability — $200/mo at 2,000 RU baseline, up to $800/mo at peak 10,000 RU.

Secrets detected (39)

Three Grafana/Swagger vars removed by denylist (#34). `MONGODB_URI` and `SPRING_DATA_MONGODB_URI` are `infra_managed` (#35) — `mongodb-url` written by the Cosmos module is authoritative.

Group	Count	Key examples
Ping Identity / OIDC	13	`PINGFED_CLIENT_SECRET`, `PING_AUTH_ISSUER_URI`, `PINGFED_DISCOVERY_URL`, `PING_CLIENT_ID`
RabbitMQ / AMQP	3	`SPRING_RABBITMQ_PASSWORD`, `AMQP_BROKER_PASSWORD`, `MQ_PASSWORD`
Redis	2	`REDIS_HOST`, `REDIS_KEY`
ELERA POS	3	`ELERA_BASE_URL`, `ELERA_SERVICE_ACCOUNT_PASSWORD`, `ELERA_REDIRECT_URI`
Kount fraud	2	`KOUNT_API_KEY`, `KOUNT_BASE_URL`
Spring Cloud Config	1	`CLOUD_CONFIG_URL`
Compliance APIs	4	`COMPLIANCE_BASE_URL`, `DB_URI`, `COM_KROGER_POS_COMPLIANCELOGGINGAPI_ENDPOINT`
Auth / crypto	2	`CIS_RSA_PUBLIC_KEY`, `AUTH_TIMEOUT_CONNECTION`
Other endpoints	9	`PING_PONG_BASE_URL`, `LOADER_CATALOG_BASE_URL`, `AUTHORIZATION_*`, etc.

> ⚠ **Warning:** High secret count (39) detected. Review whether all env vars need to be in Key Vault — consider grouping with a secrets manager pattern or filtering observability vars.

What was missed

Missing	Root cause	Impact
AMQP direct reply-to migration risk	No Migration Risk Evaluator — `spring-rabbit` → Service Bus mapping silently drops AMQP 0-9-1 / 1.0 protocol incompatibility	App may fail on `RabbitTemplate.convertSendAndReceive()` or `amq.rabbitmq.reply-to` patterns
JMX / management secondary port	Dockerfile analyzer reads only the first `EXPOSE` line	Missing `containerPort`; no NetworkPolicy rule for management port
Spring Cloud Config startup dependency	`CLOUD_CONFIG_URL` captured as a secret but not modeled as a startup dependency	Pod will fail to start if config server unreachable; no init container generated

All three gaps are non-blocking for pod scheduling but represent migration risks. They are driving use cases for the Runtime Enricher and Migration Risk Evaluator in architectural evolution issue #37.

Improvement Roadmap

Priority	Change	Status
~~High~~	~~Add MongoDB Spring Data to Java artifact classifier~~	✅ Done — #26
~~High~~	~~Add Spring AMQP / RabbitMQ to Java queue classifier~~	✅ Done — #27
~~High~~	~~Health check path from actuator / Spring Boot YAML~~	✅ Done — #28
~~High~~	~~Memory limits from Helm values / JVM -Xmx inference~~	✅ Done — #29
~~Medium~~	~~Min replicas from Helm values~~	✅ Done — #31
~~Medium~~	~~Secret filter denylist (Grafana, Swagger, Springdoc)~~	✅ Done — #34
~~High~~	~~Cosmos DB for MongoDB module (AKS + Container Apps)~~	✅ Done — #35
~~Low~~	~~Warn when secret count exceeds 30~~	✅ Done — #36
High	AMQP direct reply-to migration risk surfaced as warning	Open — #30, absorbed into #37
High	JMX / management secondary ports detected and exposed	Open — #32, absorbed into #37
High	Spring Cloud Config startup dependency modeled	Open — #33, absorbed into #37
High	Architectural evolution: Runtime Enricher, Helm Deep Reader, Migration Risk layer	Open — #37

Summary

With all fixes through issue #36 applied, YAMLess now produces a substantially more accurate infrastructure workspace for this service. The key gains since v2: health checks now probe `/actuator/health` rather than the Spring Boot default fallback; container memory is sized to the Helm-specified limit of `1152Mi` (preventing OOM

kills on the previous `512Mi` default); the Cosmos DB for MongoDB module is generated with production-tier autoscale configuration; and 3 Grafana/Swagger vars are no longer erroneously written to Key Vault.

Three migration risks remain undetected — AMQP protocol incompatibility, secondary ports, and Spring Cloud Config topology — all scoped to the Runtime Enricher and Migration Risk Evaluator in #37.

Production cost increases by $200–$800/month versus v2 (Cosmos DB autoscale), bringing the estimate to **$1,217–$1,867/month**.

Generated by YAMLess — https://github.com/zafar-ahmed-kr/YAMLess Date: 2026-11-11